CLASS : B. E. (E &TC) **SUBJECT : ESD**

EXPT. NO. : 02

Aim:

Interface LED with STM32F4 and Toggle the LED by using delay functions.

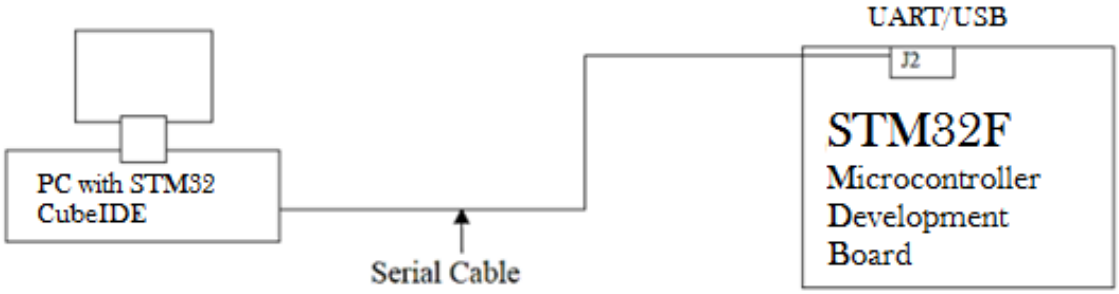
Objectives:

Create project and write code using HAL/delay function to toggle led connected to STM32F microcontroller.

Requirements:

STM32FXX Microcontroller Development Board, Personal Computer, STM32CubeIDE Software, Connecting wires, Power Supply, etc.

Block Diagram:

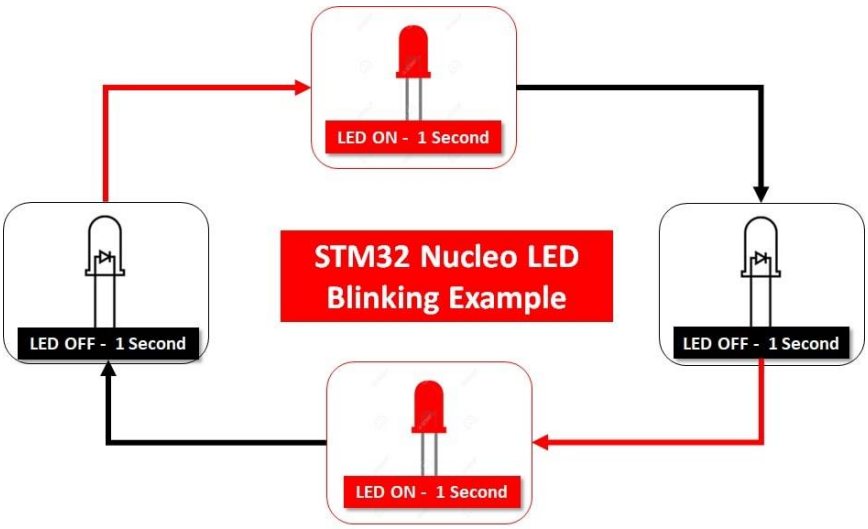


Theory:

STM32 Nucleo LED Blinking Example:

We will learn to use GPIO pins of STM32 Nucleo as digital output pins.. This STM32 board has an onboard LED with the name of LD0and it is connected with the PC_0pin.
This LED blinking example works as mentioned below:

- Onboard LED Turns on for one second which means GPIO PA_5 goes to active high state
- In next step, LED turns off for one second which means GPIO PA_5 goes to active low state
- Above two steps keep repeating as shown below:



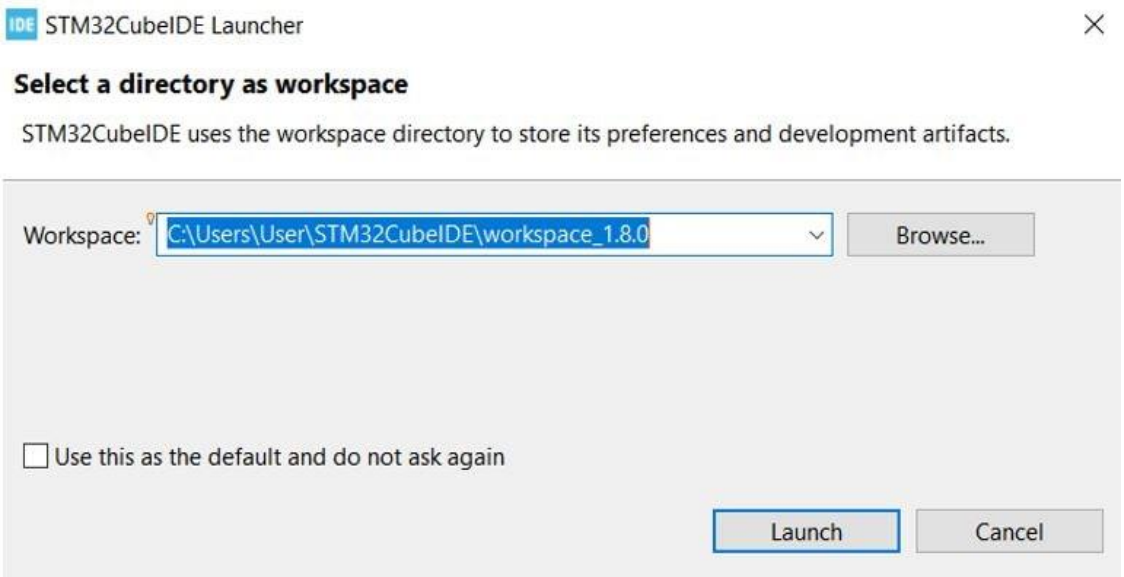
Write your first STM32 Nucleo in STM32Cube IDE:

After installing STM32CubeIDE on your Windows, Linux, or MacOS system, follow these steps to create an LED blinking project. In this section, you will learn the followings:

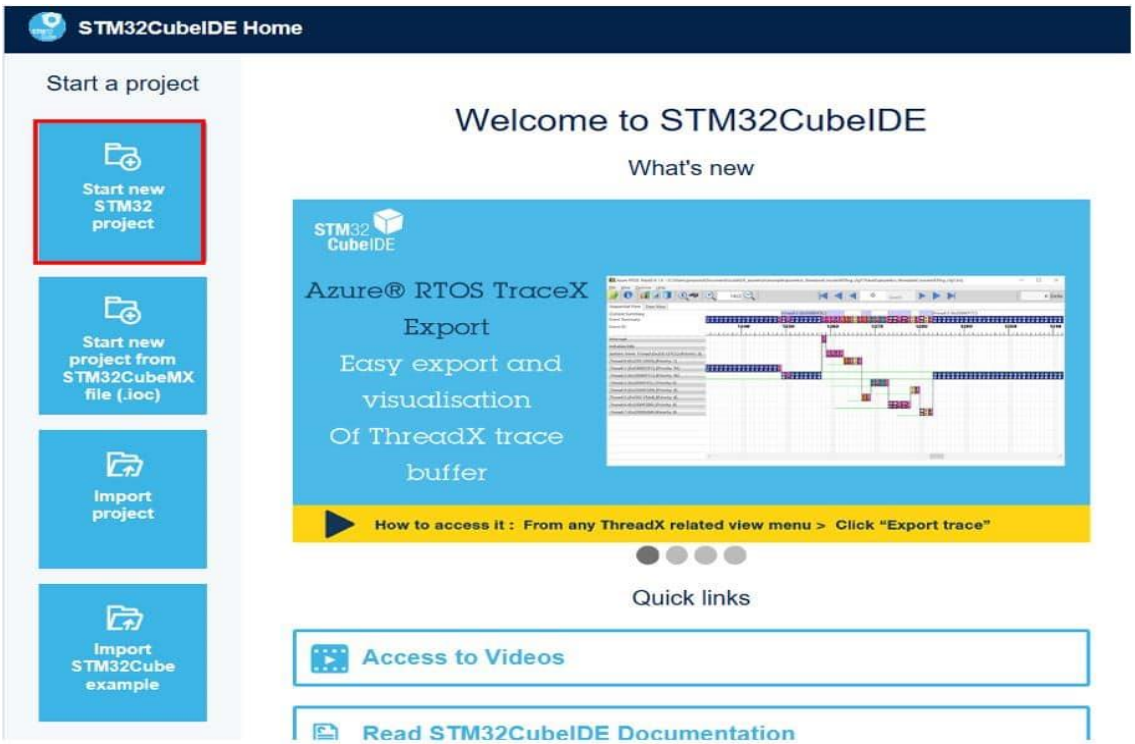
- Create and configure the STM32CubeMX project and generate the initialization code
- Program and use HAL functions to blink a LED on the NUCLEO-F103RB board.

Launch STM32CubeIDE: Open the STM32CubeIDE application on your computer. You will be asked to specify the directory for the workplace. You can specify the directory and also tick the box below to keep this as the default directory. Next, click 'Launch' to start the application.

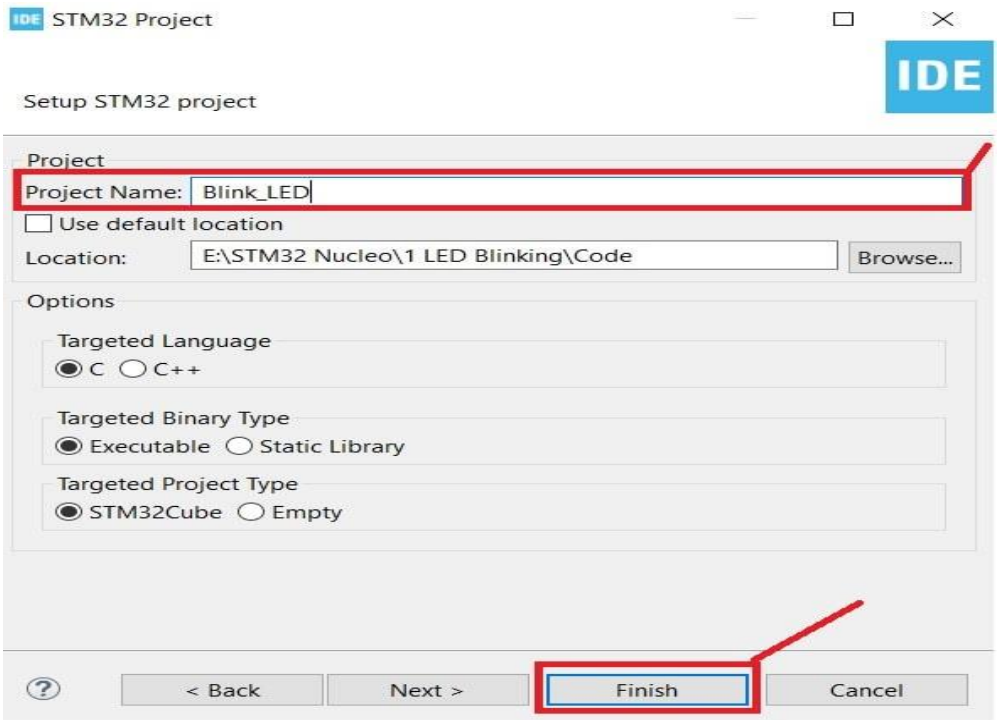
Creating STM32 Nulceo Project:



- The STM32Cube workspace will open.
- Now click ‘Start new STM32 project’.



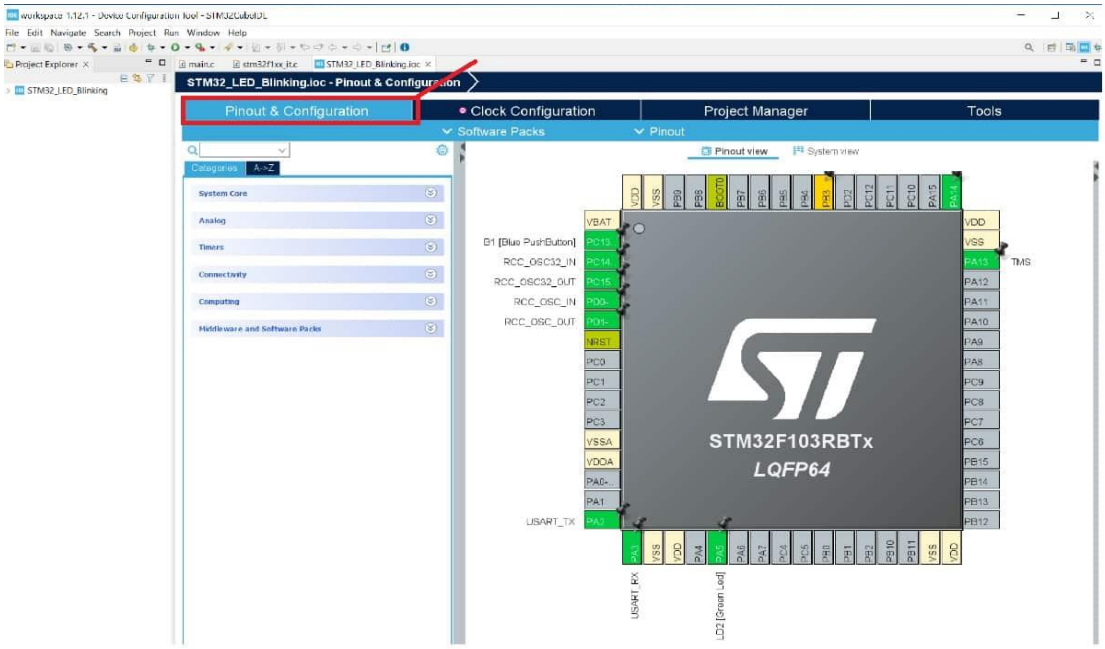
- From Board/Component Selector section, type STM32 Nucleo-Fxxxxand select it. Click next.
- Give your project a name. As we are going to blink the on-board LED of STM32 Nucleo-Fxxxx hence we have named our project ‘BLINK_LED.’ Then click ‘Finish.’



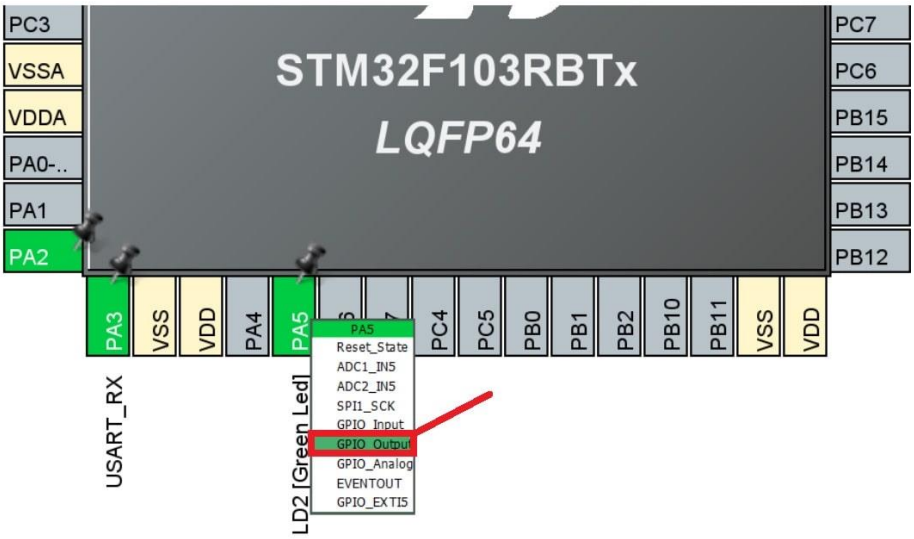
- You will be asked to open an associated perspective to start generating code with STM32CubeMX. Click 'Yes'.

Device Configuration Tool:

Now the Device Configuration Tool window will open up. In the pinout and configuration window, we can select a specific pin and its function. A single GPIO pin may have multiple alternate functions. For example, the same GPIO pin can be used for ADC, SPI, UART peripherals, etc. But only one function can be used at a time.



As mentioned earlier, we will see an LED blinking in this experiment, , we will use an onboard LED of STM32 development board. On the board LED is connected with a PC0 pin. But you can change it according to your requirement.

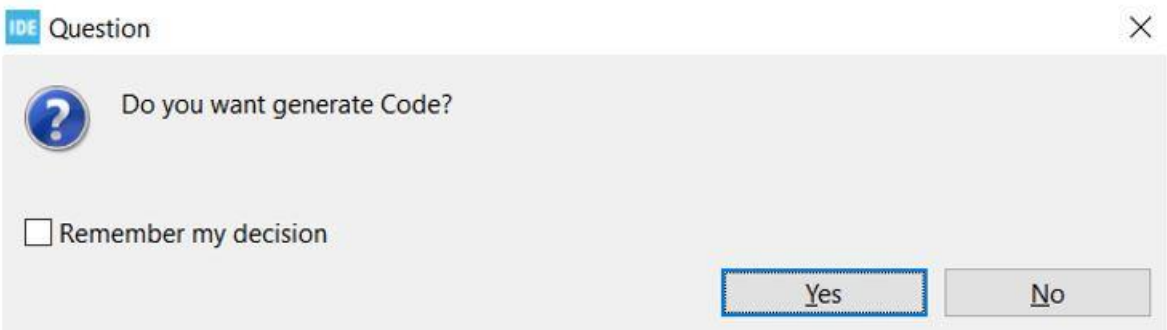


This is how it will look. Here you can see that we have attached GPIO_Output to PA5. Same should be applied to make PC0 asGPIO_Output. You can give customized names/labels to Pins.

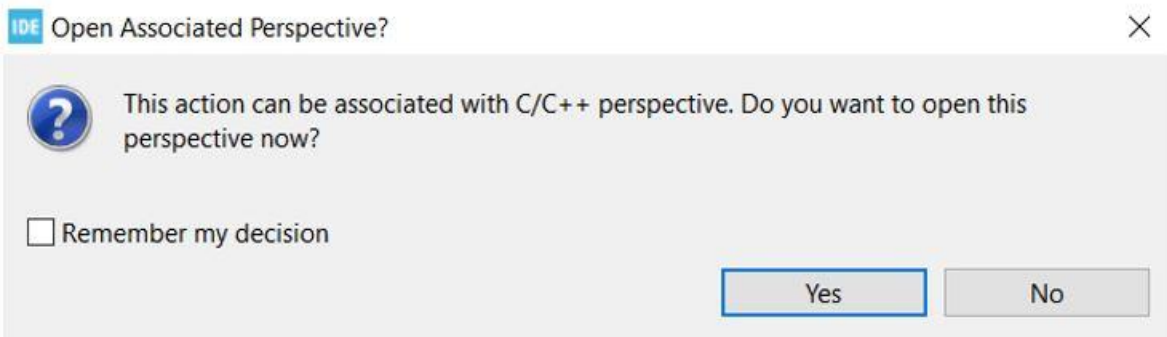
Apart from the label, there are several other options to choose from including GPIO output level, mode, pull-up/pull-down state, etc.

These configurations and parameters allow you to customize the behavior of the GPIO pins to suit the needs of your specific project and ensure proper functionality.

Now we will save our file. Press Ctrl + S. The following window will appear. Click 'Yes.' This will generate a template code for you.



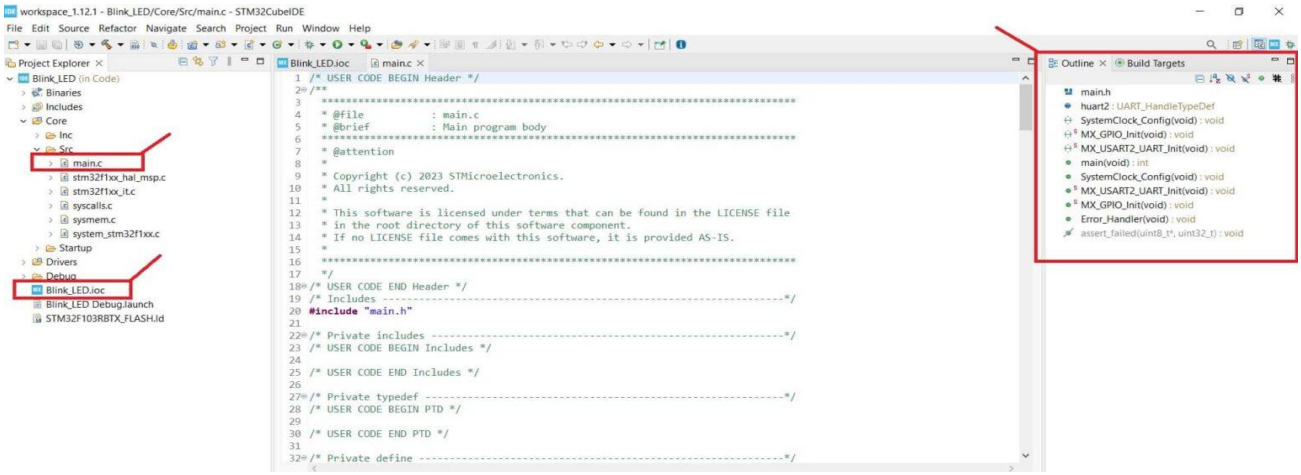
Another window will appear that will ask if you want to open the perspective. Click 'Yes.'



STM32CubeMX Generated Code:

Now the following page opens. On the right side, you will be able to view the Outline of the code. In the center, you can view the main.c file and on the left, you can view the Project Explorer. If you want to go to the Device Configuration Tool, then click the BLINK_LED.ioc tab.

STM32CubeMX will add all required HAL libraries and system level HAL libraries to use the PC0 pin of STM32 Nucleo as a digital output pin.



STM32 Nucleo LED Blinking Code:

Now let's modify the main.c file for our BLINK_LED project. Our aim is to blink the onboard LED indefinitely after a delay.

First, inside the main() function go to while(1) and insert the following lines of code. These will be responsible to blink the onboard LED infinitely.

```
while(1)
{
/* USER CODE END WHILE */
     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);// Toggle the GPIO pin state
     HAL_Delay(1000);// Delay for 1000 milliseconds
/* USER CODE BEGIN 3 */
}
```

Save the main.c file after modifying it. Now we are ready to build our project.

SMT32 GPIO HAL Driver:

The HAL driver provides functions to control the GPIO pins of the STM32 family of microcontrollers. The GPIO pins can be configured in one of the following modes:

- Digital Input mode
- Analog modeD
- Digital Output mode
- Alternate function mode

- External interrupt/event lines

Configure GPIO pins as Digital Output:

HAL GPIO library provides these functions to control, initialize and de-initialize GPIO pins.HAL_GPIO_Init() function initializes the GPIOx according to the input arguments specified to both parameters. Here the GPIOx can be (x=A, B, C…H, I) according to which GPIO port we want to use. GPIO_Init is a pointer to the structGPIO_InitTypeDef which configuration information for the specified GPIO peripheral.

```
voidHAL_GPIO_Init(GPIO_TypeDef*GPIOx,GPIO_InitTypeDef*GPIO_Init)
```

GPIO HAL Driver Toggle LED Function:

HAL GPIO driver provides toggle function HAL_GPIO_TogglePin() which can be used to toggle any GPIO pin STM32 Microcontrollers. For example, we want to toggle onboard LED of STM32 nucleo.To use this function, specified the GPIO port and its pin number to which you want to toggle.

```
voidHAL_GPIO_TogglePin(GPIO_TypeDef*GPIOx,uint16_tGPIO_Pin)
```

HAL GPIO Write Pin Function:

This function is used to set or clear a specified pin of a GPIO port. The first input argument is a PORT name, the second input argument is a pin number of the selected PORT and the third input argument is the state of the pin which can be GPIO_PIN_RESET or GPIO_PIN_SET.

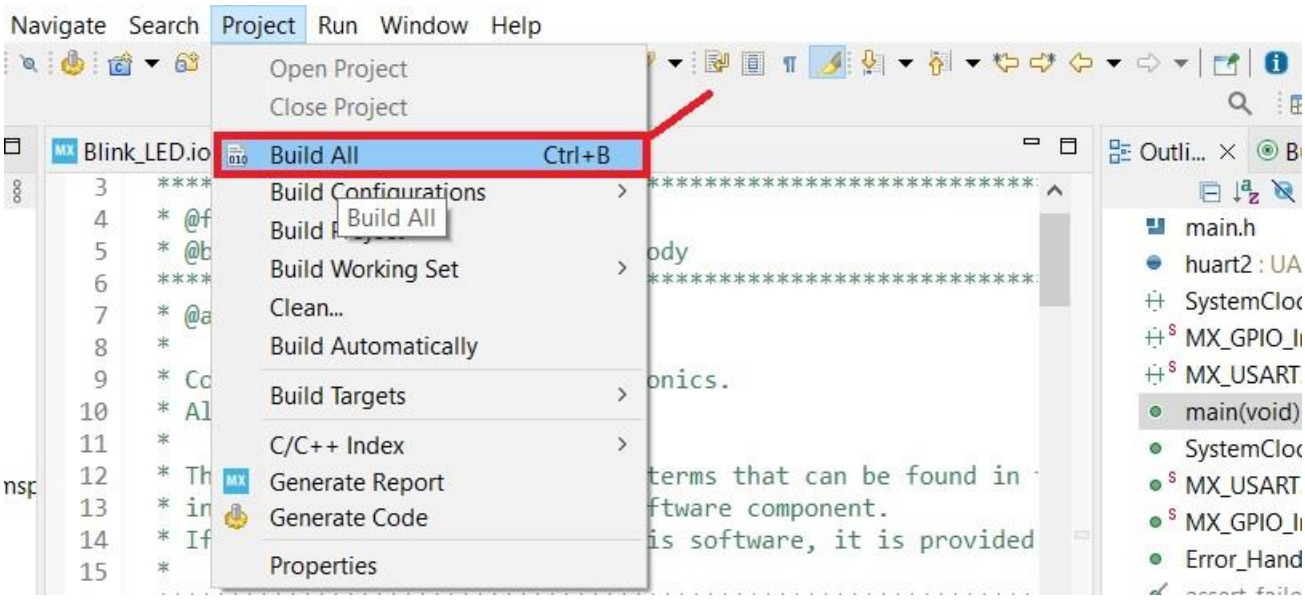
```
voidHAL_GPIO_WritePin(GPIO_TypeDef*GPIOx,uint16_tGPIO_Pin,GPIO_PinStatePinState)
```

To blink the onboard LED of STM32 Nucleo board with HAL_GPIO_WritePin() function, we can set the corresponding pins for one second and then reset the pins for one second.

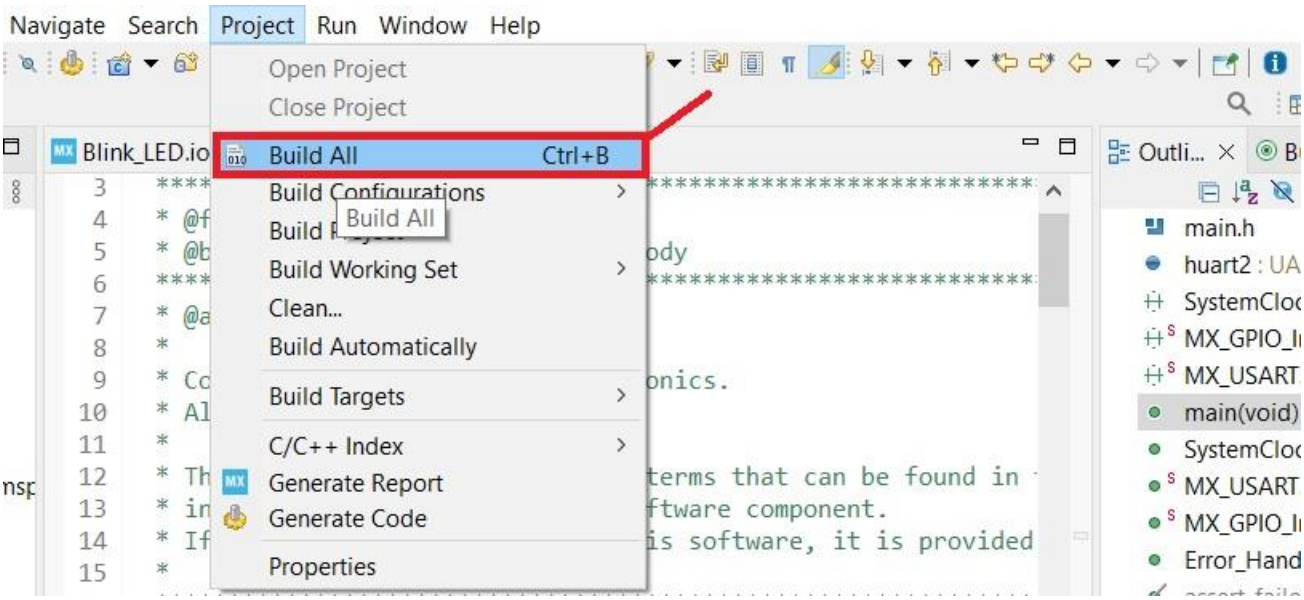
```
HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET);
Delay_ms(1000);
HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);
Delay_ms(1000);
```

Building the Project:

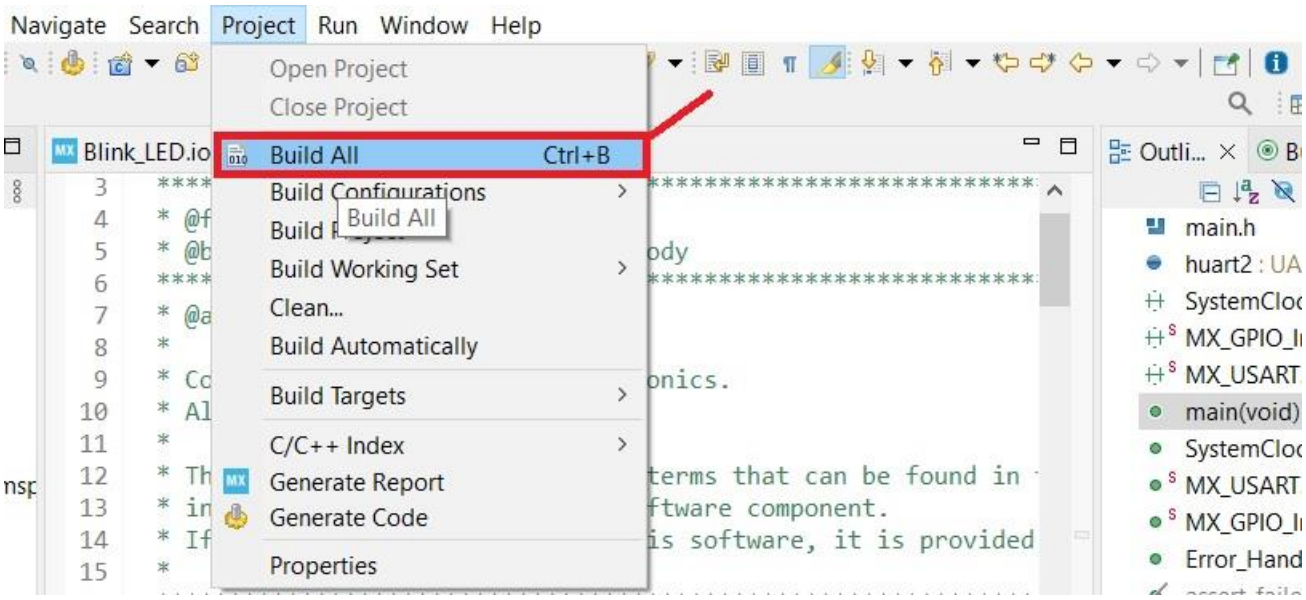
To build LED Blinking project press Ctrl + B or go to Project > Build All.



Your project will start building. After a few moments, your project will be built if there are no errors.

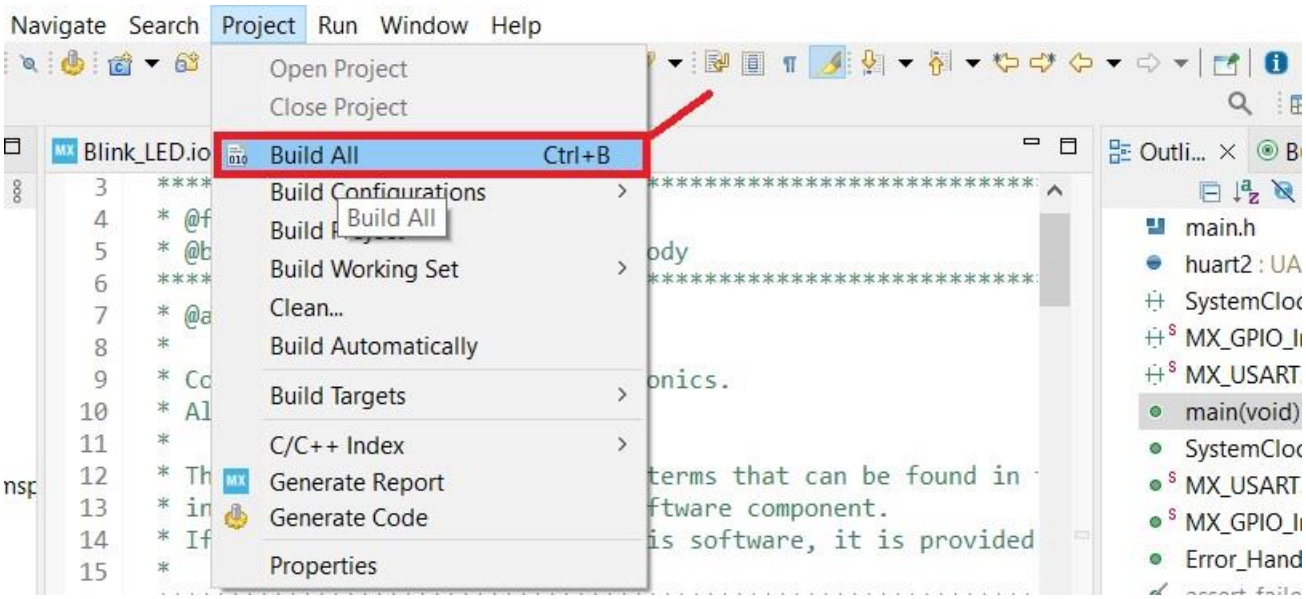


Connect STM32 Nucleo with your computer and next press the RUN button in the IDE. The following 'Edit configuration' window will open up. Click 'OK'.



After a few moments, the code will be successfully sent to the STM32 board. You can view it in the Console terminal.

Otherwise, press the RESET button on your STM32 board.



Conclusion: _____

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